

Table 4: Evaluation of G2TeXplainer across progressively enriched prompts (P1–P4) and GNN backbones. Judge-based metrics are scored from 1 (very poor) to 5 (perfect). **N. Fid.**: correctness of node importance; **Str.**: accuracy of detected motifs and node memberships; **Clar.**: coherence and interpretability of the explanation; **PW.%**: pairwise win-rate against other prompts; **Sim.**: embedding similarity across runs. The human evaluation metrics are **FacC.**: factual consistency score; **Clar.**; **Ovr.**: overall quality, a holistic judgment of usefulness; Bold indicates best and underline second best.

Prompt		LLM-as-Judge					Human Eval		
		Sim.	Str.	Clar.	N. Fid.	PW.%	FacC.	Clar.	Ovr.
GraphSAGE	GraphXAIN Adap.	–	3.754	3.862	3.881	–	<u>4.095</u>	3.143	3.190
	P1 Graph	0.712	3.576	3.284	3.687	<u>40.2</u>	4.000	3.381	2.810
	P2 1-hop	<u>0.727</u>	3.448	3.254	<u>3.697</u>	38.8	4.333	<u>4.048</u>	<u>3.476</u>
	P3 1-hop + R.	0.689	<u>3.770</u>	<u>2.994</u>	3.459	32.0	3.095	3.190	2.286
	P4 Motif-Aware + R.	0.751	4.143	4.340	3.763	89.0	4.333	4.238	3.857